

In this lesson, we examine the most common measures of dispersion: range, mean absolute deviation, variance, and standard deviation. These are all measures of **absolute dispersion**. Absolute dispersion is the amount of variability present without comparison to any reference point or benchmark.

The Range

We encountered range earlier when we discussed the construction of frequency distributions. It is the simplest of all the measures of dispersion.

Definition of Range. The **range** is the difference between the maximum and minimum values in a dataset:

$$\text{Range} = \text{Maximum value} - \text{Minimum value.} \quad (2)$$

An alternative way to report the range is to specify both the maximum and minimum values. This alternative definition provides more information as the range is reported as “from Maximum Value to Minimum Value.”

One advantage of the range is ease of computation. A disadvantage is that the range uses only two pieces of information from the distribution. It cannot tell us how the data are distributed (i.e., the shape of the distribution). Because the range is the difference between the maximum and minimum values in the dataset, it is also sensitive to extremely large or small observations (“outliers”) that may not be representative of the distribution.

Mean Absolute Deviations

Measures of dispersion can be computed using all the observations in the distribution rather than just the highest and lowest. We could compute measures of dispersion as the arithmetic average of the deviations around the mean, but the problem is that deviations around the mean always sum to 0. Therefore, we need to find a way to address the problem of negative deviations canceling out positive deviations.

One solution is to examine the absolute deviations around the mean as in the **mean absolute deviation (MAD)**.

MAD Formula. The MAD for a sample is:

$$\text{MAD} = \frac{\sum_{i=1}^n |X_i - \bar{X}|}{n}, \quad (3)$$

where $\bar{X}$ is the sample mean, n is the number of observations in the sample, and the $| |$ indicate the absolute value of what is contained within these bars.

The MAD uses all of the observations in the sample and is thus superior to the range as a measure of dispersion. One technical drawback of MAD is that it is difficult to manipulate mathematically compared with the next measure we will introduce, sample variance.

Sample Variance and Sample Standard Deviation

A second approach to the problem of positive and negative deviations canceling out is to square them. Variance and standard deviation, which are based on squared deviations, are the two most widely used measures of dispersion. **Variance** is defined as the average of the squared deviations around the mean. **Standard deviation** is the square root of the variance.